Insurance Claims Analytics & Risk Assessment

A data-driven analysis of claim outcomes, premium pricing, customer risk segmentation, and anomalous claim patterns across a 10,004-policy portfolio.

Analytics & Insights Division | Confidential | 2023-2025 Portfolio

1. Executive Summary

This report delivers a comprehensive analytical review of a 10,004-policy insurance portfolio spanning five product lines: Travel, Health, Auto, Life, and Home. The analysis covers claim outcome distribution, premium and coverage structure, customer risk profiling, anomaly detection, and policy lifecycle patterns.

The portfolio generated Rs 59.8L in premium income against Rs 600.6M in total coverage issued -- a 100x leverage ratio. Total claim exposure stands at Rs 1.69 Cr, yielding a loss ratio of 2.83x relative to premium collected. The rejection rate of 43.5% is the most operationally significant finding and the primary focus of the recommendations.

All statistical tests were conducted at a 5% significance level. Anomaly detection findings represent candidates for review, not determinate fraud indicators. Predictive model outputs are used to identify feature importance, not to automate adjudication decisions.

2. Business Problem

Three operational questions motivated this analysis:

Claim Outcomes

The rejection rate exceeds the settlement rate. Without structured analysis, the claims team cannot determine whether this reflects consistent adjudication policy or processing gaps.

Pricing Adequacy

Premium structures have not been reviewed against actual claim frequency or severity by product line. The risk that certain product lines are underpriced relative to their exposure has not been quantified.

Risk Identification

No customer-level risk tiering exists. Underwriting and renewal decisions are made without a data-driven view of which policyholders represent elevated financial exposure.

3. Data Overview

The dataset comprises 10,004 annual policies with 13 raw fields covering policyholder demographics, policy financials, and claims data. Eight additional analytical features were derived during preparation.

Dimension	Detail
Total Records	10,004
Policy Types	Travel (41.5%), Health (20.0%), Auto (15.9%), Life (12.5%), Home (10.1%)
Gender Split	Male 50.0% / Female 50.0%
Age Range	18--87 | Largest cohort: 75+ (24.3%)
Premium Range	Rs 100 -- Rs 1,099 | Mean: Rs 597
Coverage Range	Rs 10,000 -- Rs 1,10,000 | Mean: Rs 60,031
Claim Amount Range	Rs 1 -- Rs 5,499 | Mean (filed): Rs 2,994
Missing ClaimDate	4,355 records (43.5%) -- no claim filed; treated as NaT
Duplicate Records	4 removed during preparation

Limitation: The dataset does not include claim reason codes, medical diagnosis, geographic location, or customer income. Anomaly detection findings are accordingly framed as candidates for investigation, not confirmed risk events.

4. Methodology

Module	Approach	Output
Data Preparation	Schema validation, date parsing, feature engineering	cleaned_data.csv
EDA	Univariate and bivariate distribution analysis	12 executive charts
Statistical Testing	Chi-Square, ANOVA, T-Test (alpha = 0.05)	Hypothesis test results
Anomaly Detection	6-rule composite model + Isolation Forest	anomaly_flagged.csv
Risk Profiling	Weighted risk score + K-Means clustering (k=3)	risk_profiles.csv
Claim Prediction	Logistic Regression -- Settled vs Rejected	Feature importance ranking
Lifecycle Analysis	Days-to-claim timing buckets vs outcome	Lifecycle summary table

5. Claim Outcome Analysis

5.1 Overall Distribution

Of the 10,004 policies reviewed, 3,386 (33.8%) are Settled, 4,355 (43.5%) are Rejected, and 2,263 (22.6%) remain Pending. The rejection rate exceeds the settlement rate by 9.7 percentage points -- an operational imbalance that warrants structured review.

Claim Status	Count	% of Portfolio	Avg Claim Amount
Settled	3,386	33.8%	Rs 2,994
Rejected	4,355	43.5%	Rs 2,994
Pending	2,263	22.6%	Rs 3,008

5.2 Claim Status by Product Line

Rejection rates across product lines range from 42.3% (Auto) to 46.2% (Home) -- a 3.9pp spread. The consistency of this range suggests the rejection pattern is structural and not product-specific. Life policies carry the lowest settlement rate (30.9%) and warrant closer examination if adjudication guidelines differ by product.

Policy Type	Volume	Settled %	Rejected %	Pending %
Auto	1,595	35.4%	42.3%	22.4%
Health	2,000	33.8%	43.4%	22.9%
Home	1,013	32.9%	46.2%	20.9%
Life	1,248	30.9%	45.4%	23.7%
Travel	4,148	34.4%	42.9%	22.7%

5.3 Observation -- Impact -- Recommendation

Observation

The portfolio rejection rate of 43.5% is uniformly distributed across all product lines and age groups. No demographic or product-specific driver has been identified within the available data.

Business Impact

A rejection rate that exceeds the settlement rate by nearly 10 points -- without documented root cause -- creates customer experience risk, potential regulatory scrutiny risk, and limits the claims team's ability to optimise adjudication workflows.

Recommendation

Obtain rejection reason codes from the claims management system and re-run this analysis by rejection category. Until reason codes are available, the rejection rate cannot be meaningfully reduced through data-driven intervention.

6. Premium and Coverage Analysis

6.1 Premium Structure

Mean premium amounts across all five product lines fall within a Rs 4 range (Rs 595--599). ANOVA testing confirms this variation is not statistically significant. The portfolio is priced as though all product lines carry equivalent risk -- which is inconsistent with established actuarial practice.

Policy Type	Mean Premium (Rs)	Mean Coverage (Rs)	Coverage Ratio
Auto	595	60,031	142.6x
Health	597	60,031	144.4x
Home	597	60,031	143.7x
Life	598	60,031	146.8x
Travel	597	60,031	145.4x

Observation

Premium amounts are statistically indistinguishable across all five product lines. Coverage-to-premium ratios range from 142.6x to 146.8x -- a 2.9% spread -- indicating near-identical value delivery regardless of product type.

Business Impact

A flat pricing structure does not account for differences in claim frequency, claim severity, or underlying risk between product lines. Product lines with higher inherent risk (e.g. Health, Life) generate the same premium income as lower-risk lines, creating a cross-subsidisation effect that distorts portfolio profitability.

Recommendation

Engage the actuarial team to develop risk-adjusted pricing by product line. The starting point should be claim frequency and average severity per product type, cross-referenced against the current flat premium bands.

7. Statistical Analysis

Six formal hypothesis tests were conducted to assess whether claim outcomes, premium levels, and coverage ratios vary significantly across demographic and product dimensions. All tests were conducted at a 5% significance level.

Business Question	Test	Finding
Does policy type influence claim outcome?	Chi-Square	No significant association
Does gender influence claim outcome?	Chi-Square	No significant association
Do claim amounts differ by status?	ANOVA	No significant difference
Are premiums differentiated by product?	ANOVA	No significant difference
Do male and female policyholders pay different premiums?	T-Test	No significant difference
Does coverage value vary by product line?	ANOVA	No significant difference

The consistent absence of statistical significance across all six tests is a substantive finding. It indicates that the portfolio exhibits no detectable demographic or product-line bias in claim outcomes or pricing. This may reflect balanced adjudication practice, or it may be an artefact of a synthetically structured dataset. In a production environment, this finding would be validated against a longer time series and enriched with external variables.

8. Anomaly Detection

8.1 Detection Framework

Six independent detection rules were applied to identify claims exhibiting unusual financial characteristics. Records triggering three or more rules simultaneously are classified as High Risk. All flagged records are candidates for investigation -- no determination of fraud or misconduct is implied.

Detection Rule	Basis	Records Flagged
High Claim Percentile	Claim amount > 90th percentile	1,001
Isolation Forest	Multivariate anomaly (5% contamination)	501
High Claim-to-Premium Ratio	Ratio exceeds IQR upper bound	492
Early Claim Flag	Filed within 30 days of inception	455
High Utilisation	Claim / Coverage >= 80%	0
Z-Score Outlier	|Z| > 3.0 on claim amount	0

8.2 Risk Tier Summary

Risk Tier	Policies	% of Portfolio	Avg Claim (Rs)	Total Exposure (Rs)
High Risk	103	1.0%	5,121	5,27,485
Suspicious	1,714	17.1%	--	--
Normal	8,187	81.8%	--	--

Observation

103 policies trigger 3 or more anomaly rules simultaneously. Their average claim amount of Rs 5,121 is 71% above the portfolio mean of Rs 2,994. Travel policies account for the largest share (33 records), consistent with Travel's dominance of the overall portfolio.

Business Impact

Without a structured review process, high-anomaly claims may be settled without additional scrutiny, creating financial exposure from potentially overstated claims. Conversely, wrongly escalating legitimate claims carries customer experience risk.

Recommendation

Initiate a review of the 103 High Risk records using the anomaly_flagged.csv output. Prioritise by total_flags count. Establish a quarterly anomaly review cycle using this model as the triage mechanism. Refine thresholds as claim reason data becomes available.

9. Customer Risk Segmentation

9.1 Risk Scoring Model

A composite risk score was computed for each of the 10,004 policyholders using three normalised components. Weights reflect standard actuarial risk driver prioritisation.

Component	Weight	Rationale
Claim Amount	40%	Primary indicator of direct financial exposure to the insurer
Age	30%	Established actuarial proxy for claim probability and severity
Coverage Utilisation	30%	Proportion of coverage consumed -- measures concentration of exposure

9.2 Segment Distribution

Customers were segmented into three risk tiers using 33rd and 66th percentile thresholds on the composite score. K-Means clustering (k=3) was applied independently to validate the rule-based segmentation. Both methods produce broadly consistent tier assignments.

Risk Tier	Policies	% of Portfolio	Avg Risk Score	Avg Claim (Rs)
High Risk	3,402	34.0%	0.66--0.94	~4,200
Medium Risk	3,288	32.9%	0.33--0.66	~2,800
Low Risk	3,314	33.1%	0.00--0.33	~600

Observation

The portfolio is distributed near-evenly across three risk tiers, with High Risk customers (34.0%) concentrated in older age groups with above-average claim amounts and utilisation rates. K-Means clustering validates this segmentation with consistent tier assignments.

Business Impact

Without risk tiering, renewal pricing is applied uniformly regardless of customer risk profile. High Risk customers -- who generate disproportionate claim exposure -- are effectively cross-subsidised by Low Risk policyholders.

Recommendation

Integrate the risk_profiles.csv output into the renewal pricing workflow. Apply risk-adjusted premium adjustments at renewal for High Risk customers, subject to actuarial review and regulatory compliance checks. Review the top 100 highest-scoring customers (top100_high_risk.csv) at the next underwriting cycle.

10. Claim Outcome Prediction

A Logistic Regression model was trained on the 7,741 Settled and Rejected claims (Pending excluded) to identify the strongest predictors of claim outcome. The model is used as a feature importance tool -- not as an automated adjudication system.

The model achieves a cross-validated ROC-AUC of 1.0, which is atypical of real-world claims data and is noted with caution. This performance level is consistent with a structurally deterministic relationship between features and outcomes in the dataset. In production, the model should be validated against time-held-out data before deployment.

Key Predictors of Claim Outcome (by coefficient magnitude)

Claim Amount and Coverage Utilisation are the strongest predictors of whether a claim is settled or rejected. Age and Coverage Ratio follow as secondary predictors. Policy type dummies and Gender contribute minimal predictive power -- consistent with the statistical testing results in Section 7.

Practical implication: Adjudication decisions in this portfolio are most strongly associated with the financial characteristics of the claim itself, rather than the demographic profile of the policyholder.

11. Policy Lifecycle Analysis

Days-to-claim was calculated for the 5,649 policies with a filed claim. Claims were bucketed into three lifecycle stages for outcome comparison.

Claim Timing	Count	% of Claims	Avg Days	Settlement Rate
Early (0--90 days)	1,394	24.7%	46.6	59.1%
Mid (91--270 days)	2,783	49.3%	180.0	60.2%
Late (271--365 days)	1,472	26.1%	319.0	60.3%

Settlement rates are consistent across all three timing buckets (59.1%--60.3%), suggesting claim timing does not materially influence adjudication outcome within this dataset. Early claims (0--90 days) carry a marginally lower settlement rate, though the difference is not statistically significant.

Note: In live insurance portfolios, early claims -- particularly those filed within 30 days of policy inception (455 records in this dataset) -- typically warrant additional underwriting scrutiny. The absence of rejection in the early claim segment within this dataset is inconsistent with standard industry patterns and should be validated against live operational data.

12. Recommendations Summary

#	Action	Owner	Timeline
1	Review the 103 High Risk anomaly records using anomaly_flagged.csv	Claims Ops	Immediate
2	Obtain rejection reason codes and re-run claim outcome segmentation	Claims Ops	30 days
3	Actuarial review of premium pricing by product line	Underwriting	60 days
4	Establish SLA workflow for Pending claims (>60 day escalation trigger)	Claims Ops	60 days
5	Integrate risk_profiles.csv into renewal pricing workflow	Underwriting	90 days
6	Enrich dataset with claim reason codes for next analysis cycle	Data / IT	Strategic

13. Data and Model Limitations

1. Claim reason codes are absent. Root-cause analysis of rejections requires this field.
2. No geographic, income, or diagnosis data. Risk model precision is constrained accordingly.
3. The Logistic Regression ROC-AUC of 1.0 suggests structural determinism in the dataset. Production validation is required before deployment.
4. Anomaly detection flags are based on financial pattern rules only. They do not constitute evidence of fraud or policy misuse.
5. The dataset covers a single annual cohort. Trend analysis and seasonality cannot be assessed without multi-year data.